

Program 5: CI Integration using Jenkins in Docker

Step 1: Check prerequisites & create project folder

```
docker --version
docker compose version
mkdir program5
cd program5
```

Step 2: Create Dockerfile

```
nano Dockerfile
FROM jenkins/jenkins:its
USER root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

Save & exit.

Step 3: Create docker-compose.yml

```
nano docker-compose.yml
version: '3.8'
services:
jenkins:
  build: .
  container_name: jenkins-ci
  ports:
    - "8080:8080"
    - "50000:50000"
  volumes:
    - jenkins_home:/var/jenkins_home
    - /var/run/docker.sock:/var/run/docker.sock
volumes:
  Jenkins_home:
```

```
GNU nano 7.2
version: '3.8'
services:
jenkins: build: •
container_name: jenkins-ci ports:
- "8080: 8080"
- "50000:50000"
volumes:
- jenkins_home: /var/jenkins_home
- /var/run/docker.sock:/var/run/docker.sock
volumes:
jenkins_home:
```

Step 4: Build and run Jenkins

docker compose up --build

Step 5: Access Jenkins UI

Open browser:

<http://localhost:8080>



